Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd / Even Semester)

Degree, Semester & Branch: B.E, VII & ECE B
Course Code & Title: EC8701 Antennas and Microwave Engineering
Name of the Faculty member (s): S.A.Radhika

Innovative Practice Description

- **Unit / Topic: Unit …. / …Unit-V Impedance Transformation and Impedance Matching**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Team Assisted Individualization
- **Justification:**
  Impedance matching and Impedance transformation methods are quite interesting and includes more sub topics like Single Stub matching, Double stub matching and Quarter-wave Transformer. As students were already exposed to these topics in their previous semester subject Transmission lines and Waveguides, this TAI activity can make them to remember and recall what they have already learnt.
- **Time Allotted for the Activity**: 50 minutes
- **Details of the Implementation:**
  Initially the students were instructed to form a group among themselves. Around seven groups were formed with four in each. For the first 5 minutes, they were asked to select a name for their group which should be classy and highly correlated to subject. After selecting the name, they were asked to explain meaning of their group name and why they have chosen it. After that all the groups were assigned with the some questions related to topics single stub matching, Double stub matching and Quarter wave Transformer. Entire topics were divided as modules. They were allocated with 10 minutes for each module.
  They have discussed the answers and justification among their team. After completing each module, the answers for that module were discussed in common. In between the modules, to keep them engaged, activities like class poll have been conducted.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PSO3
CO5	3	2	3	3	2	2	3	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO3**	**PO8**	**PO9**	**PO10**	**PSO3**
	3	**3**	**3**	**3**	**3**	**2**
Justification for correlation	**Transmission lines fundamentals are applied**	**Design of impedance matching network**	**Group activity which requires inter-personal ethics**	**Needs individual and team work**	**effective communication is required**	**involves design and development of stub**

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    1. They can get engaged in class for the entire 50 minutes.
    2. The activity acts as a better platform for discussion.
    3. It is quite interesting**.**
  - ***Benefit of the practice:***
    1. They can easily understand while discussing and solving it as a group.
    2. Better interaction and communication due to group activity.
    3. Multiple ways to solve a problem can be obtained as all the groups are assigned with same problem statement.
  - ***Challenges faced in implementation:***
    1. To organize the activity little more time has been consumed than expected.

References:

- Team-Assisted Individualization: A Cooperative Learning Solution for Adaptive Instruction in Mathematics by Slavin, Robert E.

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Innovative Practice Description

- **Unit / Topic: Unit …. / …Unit-IV Power Divider and Magic Tee**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** Flipped Classroom
- **Justification:**

  Power divider and the magic tee are the passive microwave devices which can be understood easily. Flipped classroom practice was chosen because the topic is easy and interesting for the students to refer by themselves.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

  The students were given with the reference materials to get prepared. The students were asked to make a group among them. The questions were asked from the topic they have prepared. As it is power divider and Magic Tee, most of the questions were related to the ports. As a group, they have discussed and came up with answer for the questions. After that, they were asked to get the class back together to share the individual's group work with everyone. While discussing like this, they can get an in-depth knowledge about the topic and they can clearly express their ideologies related to that topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PSO3
CO4	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	**3**	**3**	**3**	**2**
Justification for correlation	Electromagnetic Fields fundamentals are applied	Group activity which requires inter-personal ethics	Needs individual and team work	effective communication is required	involves design and development of microwave passive devices

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

  - ***Feedback of practice from students and other stakeholders:***
    1. They can get engaged in class for the entire 50 minutes.
    2. The activity acts as a better platform for discussion.
    3. It is quite interesting.

  - ***Benefits of the practice:***
    1. They will get exposed to text books and reference materials.
    2. Better interaction and communication due to group activity.
    3. All the students can effectively take part in the activity and the idea about the topic has been reached more students.

  - ***Challenges faced in implementation:***

    1. To organize the activity little more time has been consumed than expected.

References:

- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-1 Near and Far-field region**
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen:** Mindmap

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-1 Aperture Efficiency and Effective area**
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Class poll

Signature of Faculty Member **HOD**

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-1 Link Budget and Link margin**
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** One minute paper


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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-2 Aperture antennas**
- **Course Outcome:** CO6
- **Topic Learning Outcome:** TLO17
- **Activity Chosen:** One minute paper


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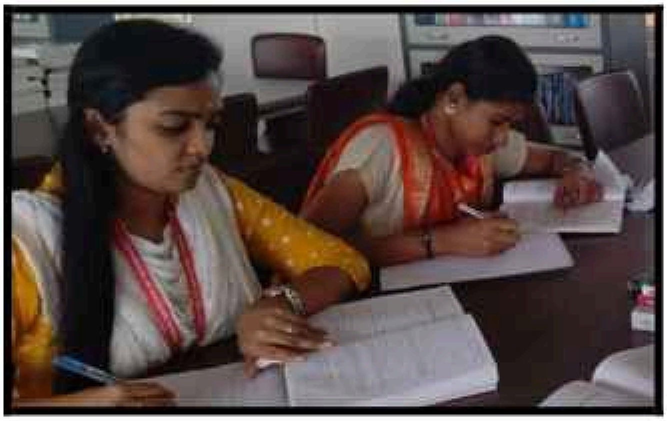
Active Learning Description

- **Unit / Topic: Unit …. / …Unit-2 Microstrip antennas**
- **Course Outcome:** CO6
- **Topic Learning Outcome:** TLO17
- **Activity Chosen:** Mind-map

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-2 Frequency Independent antennas – Design considerations and applications**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** Think pair Share

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-3 Introduction to Antenna array and Applications**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6
- **Activity Chosen:** Brain storming


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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-3 Introduction to Antenna array and Applications**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6
- **Activity Chosen:** Brain storming

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-III / Uniformly spaced arrays with non-uniform excitation amplitudes**
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Open book Test

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-IV- Power Divider, Magic Tee**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** Hardware Demonstration


Signature of Faculty Member **HOD**

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-IV- IMPATT diodes, Schottky Barrier diodes, PIN diodes**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Mind map


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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-IV- Magnetron**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Zero Minute Speech

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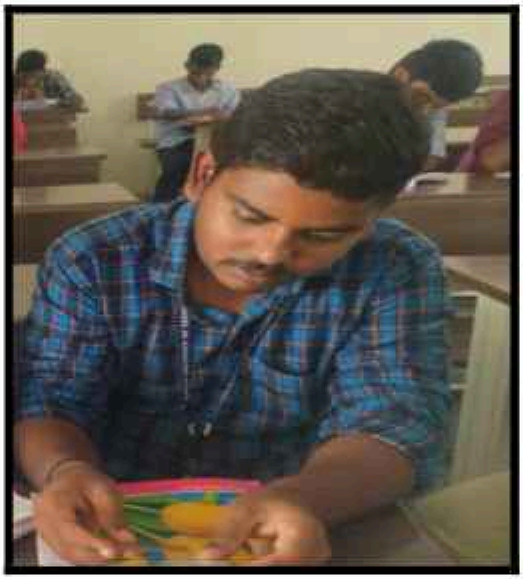
Active Learning Description

- **Unit / Topic: Unit …. / …Unit-V Microwave Filter Design**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** Hardware Demonstration

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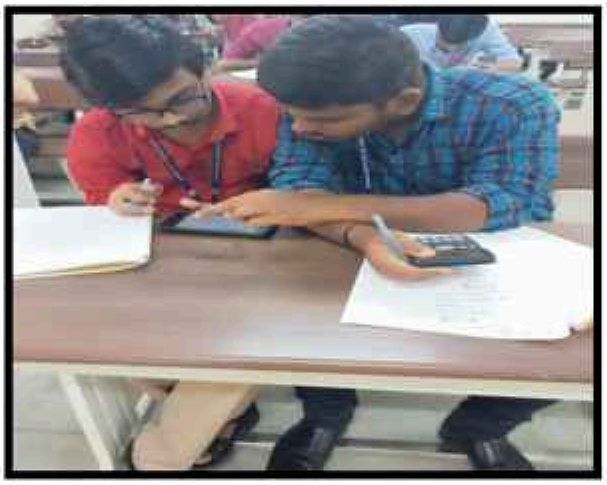
Active Learning Description

- **Unit / Topic: Unit …. / …Unit-V Low Noise Amplifier design**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** Open book Test

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Active Learning Description

- **Unit / Topic: Unit …. / …Unit-V Microwave Oscillator Design**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO14
- **Activity Chosen:** Class Poll